

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	31 January 2025
Team ID	2
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Automated standard laptop ordering using ServiceNow Flow Designer

Reference: <https://www.servicenow.com/docs>

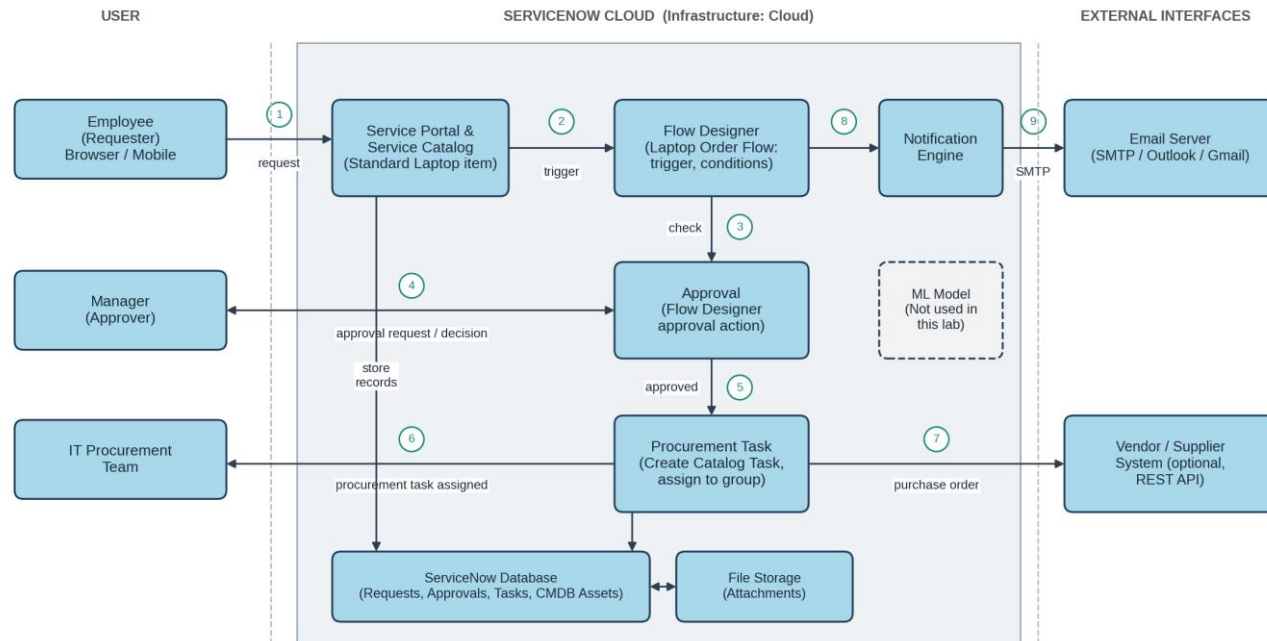


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How the employee, manager and procurement team interact with the application (Web UI on desktop / mobile browser)	ServiceNow Service Portal – HTML, CSS, JavaScript, AngularJS
2.	Application Logic-1	Request submission: Service Catalog item with form variables for the standard laptop order	ServiceNow Service Catalog, JavaScript (Client Scripts / UI Policies)
3.	Application Logic-2	Workflow automation: trigger on request, check conditions, run actions without code	ServiceNow Flow Designer
4.	Application Logic-3	Approval handling and procurement task creation after approval	Flow Designer Approval action, Create Catalog Task action
5.	Database	Stores requests, approvals, tasks, users and asset records (relational tables)	ServiceNow tables – sc_request, sc_req_item, sc_task, sysapproval_approver, alm_hardware / CMDB
6.	Cloud Database	Managed database service hosted with the ServiceNow instance	ServiceNow cloud instance database
7.	File Storage	Attachments such as quotations or approval documents	ServiceNow Attachments (sys_attachment)
8.	External API-1	Sends email notifications to employee, manager and procurement team	SMTP email / ServiceNow Notifications (Outlook, Gmail)
9.	External API-2	Optional integration to place the purchase order with the vendor / supplier system	REST API via ServiceNow IntegrationHub / REST Message
10.	Machine Learning Model	Not required for this solution (no ML used in the laptop order flow)	Not applicable
11.	Infrastructure (Server / Cloud)	Application deployment on cloud; users only need a web browser	ServiceNow Cloud (Personal Developer Instance)
	Local Server Configuration:	Not required – only a web browser and internet connection on the user's device	Local: Web browser (Chrome / Edge)
	Cloud Server Configuration:	Application hosted and managed on the ServiceNow cloud instance	Cloud: ServiceNow hosted instance

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source frameworks used by the Service Portal front end	AngularJS, Bootstrap
2.	Security Implementations	Role-based access control and ACLs, secure login, encrypted communication (HTTPS), audit of approvals	ServiceNow ACLs, Roles, HTTPS / TLS, SSO / MFA (optional)
3.	Scalable Architecture	Multi-tier cloud architecture (portal, application logic, database); flows are reusable and handle more requests without redesign	ServiceNow platform (3-tier), Flow Designer

S.No	Characteristics	Description	Technology
4.	Availability	Cloud-hosted platform with redundant infrastructure so the portal and flow are available when needed	ServiceNow cloud hosting
5.	Performance	Flow runs automatically in the background, indexed tables for quick lookup, no manual steps in the process	Flow Designer (asynchronous execution), ServiceNow caching

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>